

Extra Work

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Page 6

Monte Carlo for Pricing Options and The Big Picture

Option pricing is about valuing a contract (*option*) today by looking at many possible future outcomes of the Stocks and asking what the contract is worth on average. This is where Monte Carlo simulation enters the picture. Under a standard no-arbitrage pricing idea, the fair value of an option today is the expected payoff at maturity discounted to today. Monte Carlo is used to approximate that expectation by simulating many possible future stock paths. Additionally, for European options, closed-form pricing may be available, but for barrier, lookback and Asian options, the payoff depends on the path itself. In such cases, simulation is often the most practical approach to proceed.

The main idea

Suppose an option expires at time T and has payoff Payoff at T . If we simulate many possible future stock paths, then for each simulated path we can calculate the payoff of the option at maturity. If the simulated payoffs are $\text{Payoff}_1, \text{Payoff}_2, \dots, \text{Payoff}_N$, then the Monte Carlo estimate of the price of the option today is

$$V_0 \approx e^{-rT} \frac{1}{N} \sum_{i=1}^N \text{Payoff}_i.$$

Here: - V_0 is the price of the option today, - r is the risk-free interest rate, - T is the time of maturity, - the factor e^{-rT} discounts the future average payoff back to the present.

This discounted average payoff is interpreted as the **fair price** of the option. In other words, the price paid today for the option contract is the amount that makes the contract fair under the no-arbitrage principle. This can be understood as, what amount V_0 should one invest in bank today, where it gets compounded over time, so that at maturity T , the compounded amount $V_0 e^{rT}$ is equal to expected Payoff. This is why we need to discount the expected Payoff to today to get the Fair Price of the Option today.

Call option example: profit case

Suppose: - current stock price is $S_0 = 100$, - strike price is $K = 100$, - current option price is 8, - maturity stock price is $S_T = 130$.

For a call option, the payoff is

$$(S_T - K)^+ = (130 - 100)^+ = 30.$$

The net profit is

$$\text{Net profit} = \text{Payoff} - \text{Price of Option} = 30 - 8 = 22.$$

Here the buyer makes a profit if they exercise the option.

Call option example: loss case

Now suppose the same contract but the maturity stock price is $S_T = 90$.

Then the call payoff is

$$(S_T - K)^+ = (90 - 100)^+ = 0.$$

The net profit is

$$\text{Net profit} = 0 - 8 = -8.$$

Here the buyer may not exercise the option as, they can directly buy the stock at a lower price from the market. The loss is limited to the price of the option paid upfront to the dealer.

Page 14

Smoothness of the Simulated BM Paths by PCA

Though it may look like the simulated paths are smooth but they are actually not. It turns out one can express the j^{th} component of $\vec{a}_i$ (eigenvector of C) as proportional to sinusoidal function of j and n (if it is a n -step BM path). So, the *PCA* construction in some sense aggregates finitely many such principal components which look sinusoidal evaluated at discrete points. As because the simulated path is generating over a fine-grid and each components of the principal components look sinusoidal over these discrete time, the resulting path appears smooth. However, as one can observe from the plot adding more principal components we do get rougher paths, accounting for further explainability of the variation of the BM path.

One can find further information on this in *Glasserman, Monte Carlo Methods in Financial Engineering (sec:3.1.1)*.

Page 16-17

Brownian Bridge Construction for Barrier Crossing in a Discrete-Time Setting

Barrier options depend on whether the underlying asset price crosses a prescribed barrier during the life of the contract. In practice, stock prices are often observed only at discrete time points $0 = t_0 < t_1 < \dots < t_n = T$, but the barrier condition is naturally continuous in time. This creates a difficulty: even if the observed prices at two consecutive dates remain below the barrier, the true underlying path may still have crossed the barrier in between. The Brownian bridge construction is useful because it allows us to study this intermediate behavior, conditional on the two observed endpoint values.

Suppose we observe $S(t_i) = s_i, S(t_{i+1}) = s_{i+1}$, and assume both values lie below an upper barrier H . We are interested in the conditional probability that the actual path crosses the barrier somewhere in the interval $[t_i, t_{i+1}]$, namely

$$p_i = \mathbb{P} \left(\max_{u \in [t_i, t_{i+1}]} S(u) \geq H \mid S(t_i) = s_i, S(t_{i+1}) = s_{i+1} \right).$$

This is the natural quantity for a barrier option.

To derive this conditional crossing probability, we use the geometric Brownian motion model $dS_t = \mu S_t dt + \sigma S_t dW_t$. It is convenient to work with the log-price $X_t = \log S_t$. Over a short time interval, conditional on the endpoint values $X(t_i)$ and $X(t_{i+1})$, the process X_t behaves like a Brownian bridge. Hence, the crossing event between two observation times can be studied by simulating many Brownian bridges with the same endpoints.

Let

$$I^{(m)} = \mathbf{1} \left\{ \max_{u \in [t_i, t_{i+1}]} S^{(m)}(u) \geq H \right\},$$

where $S^{(m)}(\cdot)$ is the m -th simulated bridge between the fixed endpoints s_i and s_{i+1} . If we simulate M such independent bridges, then the sample proportion $\hat{p}_{i,M} = \frac{1}{M} \sum_{m=1}^M I^{(m)}$ is an estimator of p_i .

Since the indicators $I^{(1)}, I^{(2)}, \dots$ are *i.i.d.* Bernoulli random variables with success probability p_i , the Strong Law of Large Numbers gives $\hat{p}_{i,M} \rightarrow p_i$ almost surely as $M \rightarrow \infty$. Thus, the simulated proportion converges to the true conditional crossing probability.

In the context of a barrier option, this means that the path on $[t_i, t_{i+1}]$ can be judged probabilistically rather than only by the two observed endpoint values. If the estimated crossing probability is large, then the barrier event is likely to have occurred in that interval; if it is small, then the path likely stayed on the safe side of the barrier. Then one can construct a decision rule that if such estimated probability is large enough then the option may be exercised or not, depending on some cut-off value for the estimated probability.

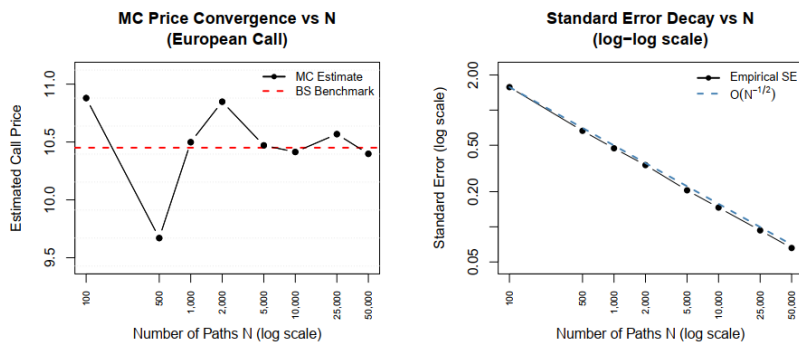
Page 23-28

Convergence with Respect to Number of Paths N

By the Law of Large Numbers the estimator $\hat{V}_N = e^{-rT} N^{-1} \sum_{i=1}^N \text{Payoff}^{(i)}$ converges to the true price as $N \rightarrow \infty$, with standard error decaying at rate $N^{-1/2}$:

$$\text{SE}(\hat{V}_N) = \frac{\sigma_{\text{payoff}}}{\sqrt{N}}$$

We study this for the European call, using `simulate_paths` (**Random Walk / Cholesky-based GBM simulation**) directly since no Section 5 wrapper exists for it.



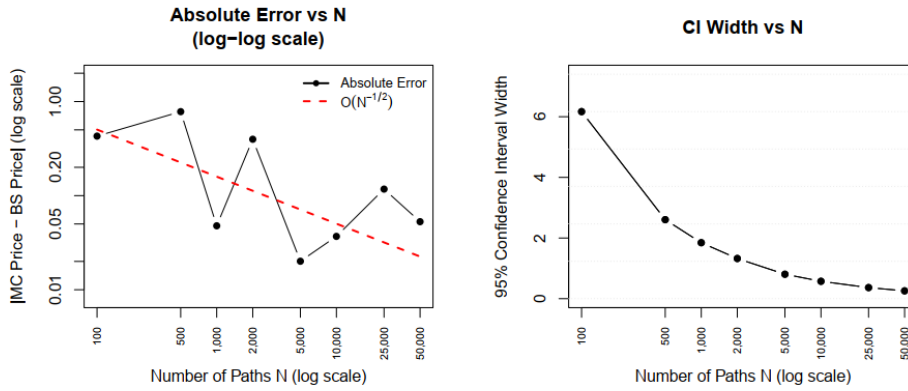
The left panel confirms that the Monte Carlo price converges toward the Black–Scholes benchmark as N grows. The right panel demonstrates the characteristic $O(N^{-1/2})$ decay of the standard error on a log–log scale, where the empirical curve closely tracks the theoretical reference line.

Error vs Benchmark and Confidence Intervals

A 95% confidence interval for the Monte Carlo estimator is constructed as

$$\hat{V}_N \pm 1.96 \cdot \frac{\hat{\sigma}}{\sqrt{N}}$$

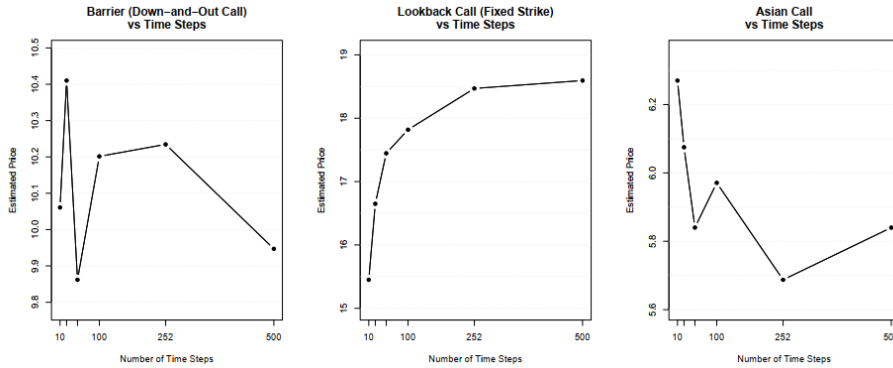
##	N	Price	SE	CI_low	CI_high	Width	Abs_Err
## 1	100	10.8795	1.57349	7.7954	13.9635	6.16809	0.42891
## 2	500	9.6699	0.66452	8.3675	10.9724	2.60490	0.78064
## 3	1000	10.4984	0.47085	9.5755	11.4212	1.84573	0.04778
## 4	2000	10.8479	0.33788	10.1857	11.5102	1.32449	0.39732
## 5	5000	10.4707	0.20554	10.0678	10.8735	0.80571	0.02009
## 6	10000	10.4137	0.14623	10.1271	10.7003	0.57321	0.03690
## 7	25000	10.5679	0.09326	10.3851	10.7507	0.36559	0.11732
## 8	50000	10.3977	0.06586	10.2686	10.5268	0.25818	0.05286



Both the absolute error and CI width follow the $O(N^{-1/2})$ envelope closely. At $N = 50,000$ the interval width is negligibly small, confirming that the simulation is practically unbiased relative to the analytical benchmark.

Convergence with Respect to Time Discretization (Steps)

Path-dependent options are sensitive to the number of time steps: coarser grids miss barrier crossings and misrepresent path extremes. We call the Section 5 pricing functions directly, varying only **steps**.



All three prices stabilize as the number of steps increases. The barrier option shows the steepest sensitivity at low step counts: a coarse grid fails to detect intra-period barrier breaches, causing the price to be overestimated. The lookback and Asian options exhibit milder but consistent step-size dependence, reflecting the smoother averaging nature of their payoffs.